Aaditya Baranwal

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Education

University of Central Florida — Orlando, FL
Ph.D. Computer Science — *Aug. 2024 – Present*

Indian Institute of Technology — Jodhpur, Rajasthan
Bachelor of Technology in Electrical Engineering, Specialization in Visual Computing — *Nov. 2020 – Jun. 2024*

Sant Atulanand Convent School — Varanasi, UP
Senior Secondary Certificate — *Aug. 2018 – May 2020*

St. John's Convent School — Varanasi, UP
Higher Secondary Certificate — *May 2016 – May 2018*

Key Experiences

Data Science Researcher — Jan. 2025 – Present
Siemens Energy — *Orlando, FL*

- Developed scalable zero-shot hazard detection tools for industrial sites using Foundation Models and VLMs while adapting models to overcome limited industrial image representation in pretraining.

Artificial Intelligence Engineer — Jun. 2024 – Aug.2024
AlphaBake — *Bengaluru, IN*

- Creating scalable stable diffusion workflows for virtual try-on. Improvised realism in the garment-fitting pipeline by inculcating IP adapters and training garment LoRAs.

Deep Learning Research Intern — Jun. 2023 – Nov. 2023
Bosch Global Software Technologies — *Bengaluru, IN*

- Analysis and finetuning of Foundation Models for Autonomous Driving including Detection and Segmentation.
- Empowering the Segment Anything Model to interpret prompts of text modality, harnessing the transformative capabilities of a Visual GPT-inspired architecture to enable interactive segmentation and uncertainty estimation.

Computer Vision Research Intern — Jul. 2022 – Sep. 2022
Bosch Global Software Technologies — *Remote, IN*

- Established an optimized pipeline of the SOTA Multi-Object Tracking algorithms. Accelerated the boundary-based LSTM for trajectory determination using CUDA.
- Explored ways to tackle label swapping and occlusion problems during interference on KITTI.

Computer Vision Intern — Jun. 2022 – Aug. 2022
Dimensionless Technologies — *Mumbai, IN*

- Built a scalable pipeline using EfficientNets and Residual Attention for commodity image discrimination and developed a counterfeit detection AI for e-commerce, trained on HPCs with CUDA.

Python Developer — Jan. 2021 – Apr. 2021
Mathrithms — *Rajasthan, IN*

- Extending BunnyCDN's cloud and storage services, creating a Python SDK, allowing functionalities of BunnyCDN apiary through Python, and achieved a total count of 75000+ global downloads from PYPI.

Research and Teaching

Student Researcher — Aug. 2023 – Jan. 2024
Image Analysis and Biometrics Lab — *Jodhpur, IN*

- Developed advanced facial recognition paradigm using Vision Transformers and multimodal learning, achieving 68% accuracy in zero-shot face verification with enhanced system interpretability and adaptability.

Teaching Assistant - Signals and Systems — Aug. 2023 – Dec. 2023
Indian Institute of Technology — *Jodhpur, IN*

- Assisting Dr. Himanshu Kumar in teaching EEL2010 - Signals and Systems to 320 students, covering Signal Theory (Beginner-Intermediate) and Linear Algebra (Intermediate-Advanced). Responsibilities include tutoring, evaluation, setting curated standards for questions, and teaching the MATLAB portion of the course.

Key Projects

Transformer Based APR - ViTLoc | *Pytorch, Pytorch3D, Timm* Mar. 2024 – May 2024

- Achieved SOTA Absolute Pose Regression with two novel approaches ($1.78m/3.37°$ & $2.02m/3.18°$).
- ThreeFormer employs a multi-stream approach with dedicated position, orientation, and backbone transformers to emphasize different visual cues, using self-attention to aggregate pose-relevant features. FeatFormer integrates 3D feature matching. ViT backbone to extract dense visual features projected into a 3D feature volume via NeRF.

Gradient-Based Enhanced Adverserial Robustness - GEAR | *Pytorch, Tmm* Mar. 2024 – May 2024

- Developed a robust GAN-generated image detection method using an ensemble of CNN and ViT gradients with depthwise convolutions. Achieved state-of-the-art results: cross-model accuracy $> 75\%$, cross-category accuracy $> 89\%$, and overall accuracy of 97.75%, demonstrating resilience against image and frequency space artifacts.

Retrieval Augmented Math QA | *PyTorch, HuggingFace* Jan. 2023 – Apr. 2023

- Devised a novel methodology for Math QA using MWP-BERT as the solver and a custom Encoder/Retriever. Implemented PLM Encoding for Word Problem matrix representation with self-supervised and fully-supervised pre-training objectives.

Gesture Controlled AMR - VOWEL | *ROS, TensorFlow, PyTorch, Gazebo* Mar 2022 – Aug. 2022

- Designed and implemented a vision-based semi-autonomous mining robot using ROS incorporating instance segmentation, hand detection (YOLO & Keypoint Mapping), and real-time gesture action classification (LSTM) with path planning to address challenges in the mining industry.

DRDO's UAV Guided UGV | *OpenCV, ROS, Shell, Linux* Feb. 2022 – Apr. 2022

- Designed a UAV travel algorithm employing K-means segmentation and Depth-Vision for trajectory construction, with stable path-following using PID control and Differential Kinetics. Developed a computation-efficient pipeline for ROS code, optimizing image processing and robot control mechanisms.

Achievements

GRE: 320 (170(Math) + 150(Verbal) 2023

Winner Michelin Mobility Innovation Challenge 2022

Ranked 7th among 23 participating IITs (Techmeet 10.0 DRDO UAV) 2022

Bagged a Perfect Score of 100/100 ABU Robocon' 22 2022

Ranked 4th in Samsung NLP challenge(Kaggle) 2022

All India Rank-65 ISI Entrance Test 2020

Skills

Programming Languages | *Python, C++, Matlab, Unity, SQL, HTML, CSS, JavaScript, Shell*

Tools & Utilities | *HF, Langchain, Weights&Biases, GCP, AWS, Git, Docker, LaTeX, OpenAI-Gym*

Libraries | *PyTorch, TensorFlow, OpenCV, SkLearn, Numpy, Pandas, etc.*

Theoretical & Applied | *Deep and Reinforcement Learning, Stable Diffusion, Computer Vision, NLP/NLU, Robotics*

Positions of Responsibility

Festival Cheif of Aaftaab The Literary Fest - IIT Jodhpur Feb. 2024

- Headed the organization and orchestration of 3rd ever Aaftaab, handling a team of 50+ members over multiple verticals, 500+ physical participants, guests, and multiple sponsors in various competitions and events.

Founder and Head of Artificial Intelligence and Data Society - IIT Jodhpur 2022-2023

- Founded RAID, an AI-focused community, orchestrating workshops, mentoring projects, and fostering a collaborative environment, thereby ensuring continual professional growth for members.

Secretary of Literature Society - IIT Jodhpur 2022-2023

- Leading a diverse, five-club Literature Society, I champion literary exploration through curated events and workshops, empowering students to delve into their literary passions.